FB AI for 2D to 3D
FB also developed technology to convert 2D photos to 3D, and wants to use it on photos of puppies. **https://dgit.in/apr20-73**

Microsoft 3D generators
Microsoft's approach for converting 2D to 3D involves using generative adversarial networks. **https://dgit.in/apr20-74**

Generating 3D Models from 2D Images with Nvidia DIB-R

Nvidia, along with other researchers, have developed an AI system to create realistic 3D models from simple 2D images using differentiable rendering

Tabitha Thomas | feedback@digit.in

Imagine if you could build a game like GTA or Skyrimm just from pictures. You click a picture, run it through a software and out comes a 3D model. Basically, take any 2D image and convert it into a 3D model. As simple as that. Creating 3D worlds would no longer be the sole domain of an army of coders and designers. You can create and run around in GTA <Your City>.

If, like us, you are also drooling over this, thank Nvidia for making it possible. Nvidia researchers along with researchers from Vector Institute, University of Toronto and Aalto University recently developed an AI system, called DIB-R, capable of creating a complete 3D model from any 2D image.

NVIDIA DIB-R MODEL

The AI model is based on how we perceive 3D images. If you think the human eye "sees" images in 3D, you are wrong. We perceive images in 2D i.e. we see surfaces, objects, and structures. Our brain then adds depth to these images to produce a 3D effect.

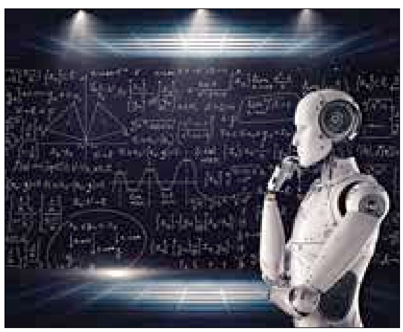
The hunt was on to create an architectural framework that could do this with the seamless integration of machine learning techniques. Researchers at NVIDIA attempted doing this by using a technology called Differential Interpolation Based Renderer or DIB-R. By using DIB-R, the researchers were able to render a fully-textured 3D model from just one 2D photograph and the whole process took just a few seconds. This high-fidelity rendering was the result of an encoder-decoder architecture which was responsible for transforming the input (in this case, a 2D image) into a vector map that could predict the texture, shape, colour, depth, lighting etc of the image.

TRAINING THE MACHINE:

In order to get the desired result, this neural network first needed a certain amount of training. In order to train it, vast amounts of data were fed into it. These included the likes of ready 3D models, images that were later transformed into 3D models as well as collated bunch of photographs of the same object from different angles. After over 48 hours of processing these images, the AI was equipped to generate 3D models from images that it had not analysed previously. All of this was now being done by the machine, minus any intervention from the researchers.

The machine was now capable of predicting what a 2D image would look like when in 3D. It would then create a 3D model of the image by taking into consideration aspects such as texture, lighting, depth etc. To do this, it takes the help of traditional templates of 3D shapes (like polygon spheres) and makes the necessary alterations to match it to the shape of the object in the 2D image.

During the first round of data input, one of the database sets used was a collection of bird images. After the DIB-R had completed its training, the model was seen to be capable of taking images of birds and replicating



A large amount of data was fed to the neural network about 3D models, images made from 3D models, etc.